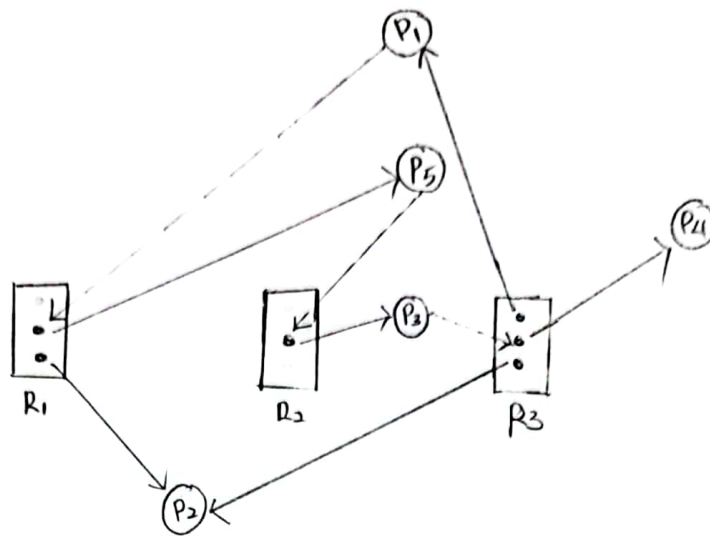


Question 1

a) i)



ii) The RAG has one cycle:

$\underline{P_1} \rightarrow R_1 \rightarrow \underline{P_5} \rightarrow R_2 \rightarrow \underline{P_3} \rightarrow R_3 \rightarrow \underline{P_1}$

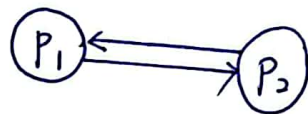
P_1, P_3, P_5 are in a deadlock state.

iii)

this system is not deadlock

when P_4 releases R_3 , this resource can be assigned to P_3 .

b)



A cycle from P_1 to P_2 and again to P_1 .

The system is in a deadlock state.

Allocation					MAX				Need			
	A	B	C	D	A	B	C	D	A	B	C	D
P ₀	2	0	1	0	5	0	5	2	3	0	4	2
P ₁	2	1	0	1	2	3	1	5	0	2	1	4
P ₂	3	2	1	2	5	3	2	3	2	1	1	1
P ₃	0	3	2	1	2	5	5	4	2	2	3	3

P₃ Request (1, 2, 0, 0) ≤ Available (3, 3, 2, 1)

Request₃ ≤ Needs₃

$$\text{Available} = (3, 3, 2, 1) - (1, 2, 0, 0) \\ = (2, 1, 2, 1)$$

$$\text{Allocation}(P_3) = (0, 3, 2, 1) + (1, 2, 0, 0) \\ = (1, 5, 2, 1)$$

$$\text{Need}(P_3) = (2, 2, 3, 3) + (1, 2, 0, 0) \\ = (3, 4, 3, 3)$$

Allocation					MAX				Need			
	A	B	C	D	A	B	C	D	A	B	C	D
P ₀	2	0	1	0	5	0	5	2	3	0	4	2
P ₁	2	1	0	1	2	3	1	5	0	2	1	4
P ₂	3	2	1	2	5	3	2	3	2	1	1	1
P ₃	1	5	2	1	2	5	5	4	3	4	3	3

1. Work = Available (2, 1, 2, 1)

2. Need(P₂) ≤ Work

$$(2, 1, 1, 1) \leq (2, 1, 2, 1)$$

$$\text{Work} = (2, 1, 2, 1) + (3, 2, 1, 2) \\ = (5, 3, 3, 3)$$

3. Need(P₁) ≤ Work

$$(0, 2, 1, 4) \leq (5, 3, 3, 3)$$

$$\text{Work} = (5, 3, 3, 3) + (2, 1, 0, 1) \\ = (7, 4, 3, 4)$$

4. Need(P₃) ≤ Work

$$(3, 4, 3, 3) \leq (7, 4, 3, 4)$$

$$\text{Work} = (7, 4, 3, 4) + (1, 5, 2, 1) \\ = (8, 9, 5, 5)$$

5. Need(P₀) ≤ Work

$$(3, 0, 4, 2) \leq (8, 9, 5, 5)$$

$$\text{Work} = (8, 9, 5, 5) + (2, 0, 1, 0) \\ = (10, 9, 6, 5)$$

Safe sequence

P₂ → P₁ → P₃ → P₀

(i) Total instances

$$A: 2 + 2 + 3 + 1 = 8$$

$$B: 0 + 1 + 2 + 5 = 8$$

$$C: 1 + 0 + 1 + 2 = 4$$

$$D: 0 + 1 + 2 + 1 = 4$$

Question 2.

a)

$$\text{Page size} = \text{frame size} = 2^8$$

$$\text{No. of frame} = 1024$$

$$\text{No. of page} = 512 \text{ pages.}$$

i)

$$\begin{aligned} \text{Total size of the logical address space} &= \text{total pages} \times \text{page size} \\ &= 512 \times 2^8 \\ &= 131072 \text{ bytes.} \end{aligned}$$

ii)

$$\begin{aligned} \text{Total size of the physical address space} &= \text{total frames} \times \text{frame size} \\ &= 1024 \times 2^8 \\ &= 262144 \text{ bytes.} \end{aligned}$$

iii)

$$\text{Page size} = \text{frame size} = 2^8 = 256$$

iv.)

$$\text{No. of frames} = 1024 = 2^{10}$$

v.)

$$\begin{aligned} \text{Number of entry} &= \text{number of pages} \\ &= 512 \text{ entries.} \end{aligned}$$

vi)

$$\text{offset bit} = 8 \text{ bits.}$$

$$\begin{aligned} \text{Logical memory} &= 131072 \\ &= 2^{17} \end{aligned}$$

$$\begin{aligned} \text{Physical memory} &= 262144 \\ &= 2^{18} \end{aligned}$$

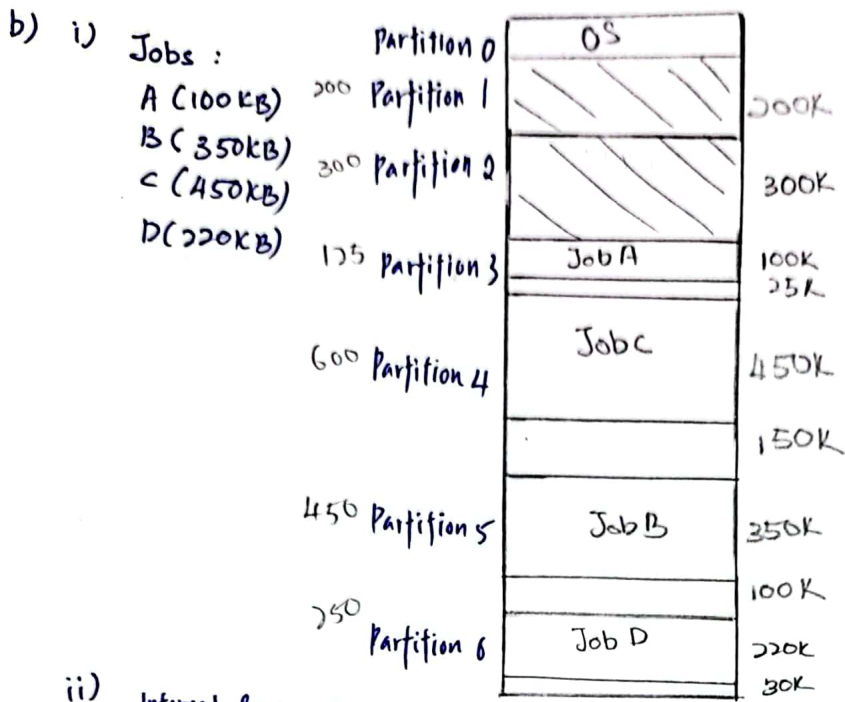
$$\begin{aligned} \text{Page number bit} &= 17 - 8 \\ &= 9 \text{ bits.} \end{aligned}$$

$$\begin{aligned} \text{Frame number bit} &= 18 - 8 \\ &= 10 \text{ bits} \end{aligned}$$

vii)

$$\begin{aligned} \text{frame \# : frame offset} \\ 1000 : 0100 0010 &\rightarrow 8 : 66 \\ \text{logical address} &= \text{page \#} \times \text{page size} + \text{offset} \\ &= 3 \times 2^8 + 66 \\ &= 834 \end{aligned}$$

Question 2(b)



ii) Internal fragmentation occur.

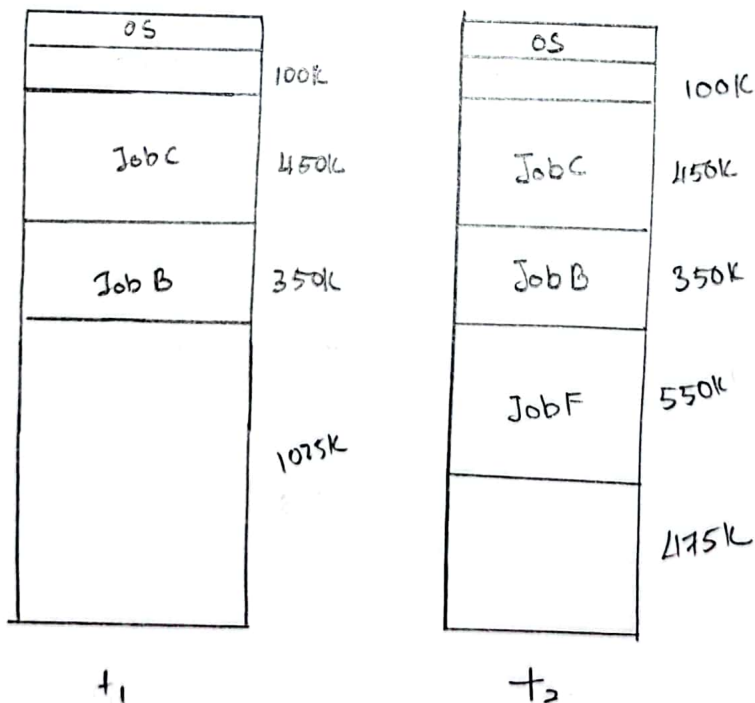
Yes.

Size of internal fragmentation

$$= 25K + 150K + 30K$$

$$= 305K$$

c)



i) t_0 : 1120K and t_1 = 800K

ii) t_1 = 800K

iii) t_1 = 800K

iv) t_2 = 1350K